```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, classification_report, accuracy_s

# Load and clean dataset
df = pd.read_csv("617ec7a0-b7f1-423e-b810-23f59803ffb6.csv")

# Handle missing values
df['Arrival Delay in Minutes'] = df['Arrival Delay in Minutes'].fillna(df['Arri

# Feature Engineering
service_cols = ['Seat comfort', 'Inflight wifi service', 'Inflight entertainmer
                'Online support', 'Ease of Online booking', 'On-board service',
                'Leg room service', 'Baggage handling', 'Checkin service',
                'Cleanliness', 'Online boarding']
df['Service_Quality_Avg'] = df[service_cols].mean(axis=1)
df['Total Delay'] = df['Departure Delay in Minutes'] + df['Arrival Delay in Mir
df['Had_Delay'] = (df['Total Delay'] > 0).astype(int)

# Encode categorical variables
categorical_cols = ['Customer Type', 'Type of Travel', 'Class']
for col in categorical_cols:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col].astype(str))

# Encode target variable
target_encoder = LabelEncoder()
df['satisfaction_encoded'] = target_encoder.fit_transform(df['satisfaction'])

# Define features
features = ['Customer Type', 'Age', 'Type of Travel', 'Class', 'Flight Distance
            'Seat comfort', 'Departure/Arrival time convenient', 'Food and drir
            'Gate location', 'Inflight wifi service', 'Inflight entertainment',
            'Online support', 'Ease of Online booking', 'On-board service',
            'Leg room service', 'Baggage handling', 'Checkin service',
            'Cleanliness', 'Online boarding', 'Total Delay', 'Had_Delay',
            'Service_Quality_Avg']

X = df[features]
y = df['satisfaction_encoded']

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, randon
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# Scale features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Train Logistic Regression
log_reg = LogisticRegression(C=1.0, solver='lbfgs', max_iter=1000, class_weight
log_reg.fit(X_train_scaled, y_train)

# Predictions
y_pred = log_reg.predict(X_test_scaled)
y_pred_proba = log_reg.predict_proba(X_test_scaled)[:, 1]

# Calculate metrics
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
roc_auc = roc_auc_score(y_test, y_pred_proba)

print("="*50)
print("MODEL PERFORMANCE METRICS")
print("="*50)
print(f"Accuracy:  {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall:    {recall:.4f}")
print(f"F1-Score:  {f1:.4f}")
print(f"ROC-AUC:   {roc_auc:.4f}")
print("="*50)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)

# Visualize Confusion Matrix
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Dissatisfied', 'Satisfied'],
            yticklabels=['Dissatisfied', 'Satisfied'])
plt.title('Confusion Matrix - Logistic Regression')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()

# Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=['Dissatisfied', 'Sati

# ROC Curve
fpr, tpr, thresholds = roc_curve(y_test, y_pred_proba)
```

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plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, label=f'Logistic Regression (AUC = {roc_auc:.3f})', linewidt
plt.plot([0, 1], [0, 1], 'k--', label='Random Classifier', linewidth=1)
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve - Airline Satisfaction Prediction')
plt.legend()
plt.grid(alpha=0.3)
plt.show()

# Feature Importance
coefficients = log_reg.coef_[0]
coef_df = pd.DataFrame({
    'Feature': features,
    'Coefficient': coefficients,
    'Odds_Ratio': np.exp(coefficients)
})
coef_df['Abs_Coefficient'] = np.abs(coef_df['Coefficient'])
coef_df = coef_df.sort_values('Abs_Coefficient', ascending=False)

print("\n" + "="*50)
print("TOP 10 FEATURES INFLUENCING SATISFACTION")
print("="*50)
print(coef_df[['Feature', 'Coefficient', 'Odds_Ratio']].head(10).to_string(inde

# Visualize top coefficients
plt.figure(figsize=(10, 8))
top_features = coef_df.head(15)
colors = ['green' if x > 0 else 'red' for x in top_features['Coefficient']]
plt.barh(top_features['Feature'], top_features['Coefficient'], color=colors)
plt.xlabel('Coefficient Value')
plt.title('Top 15 Features Influencing Satisfaction')
plt.axvline(x=0, color='black', linestyle='--', alpha=0.5)
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```

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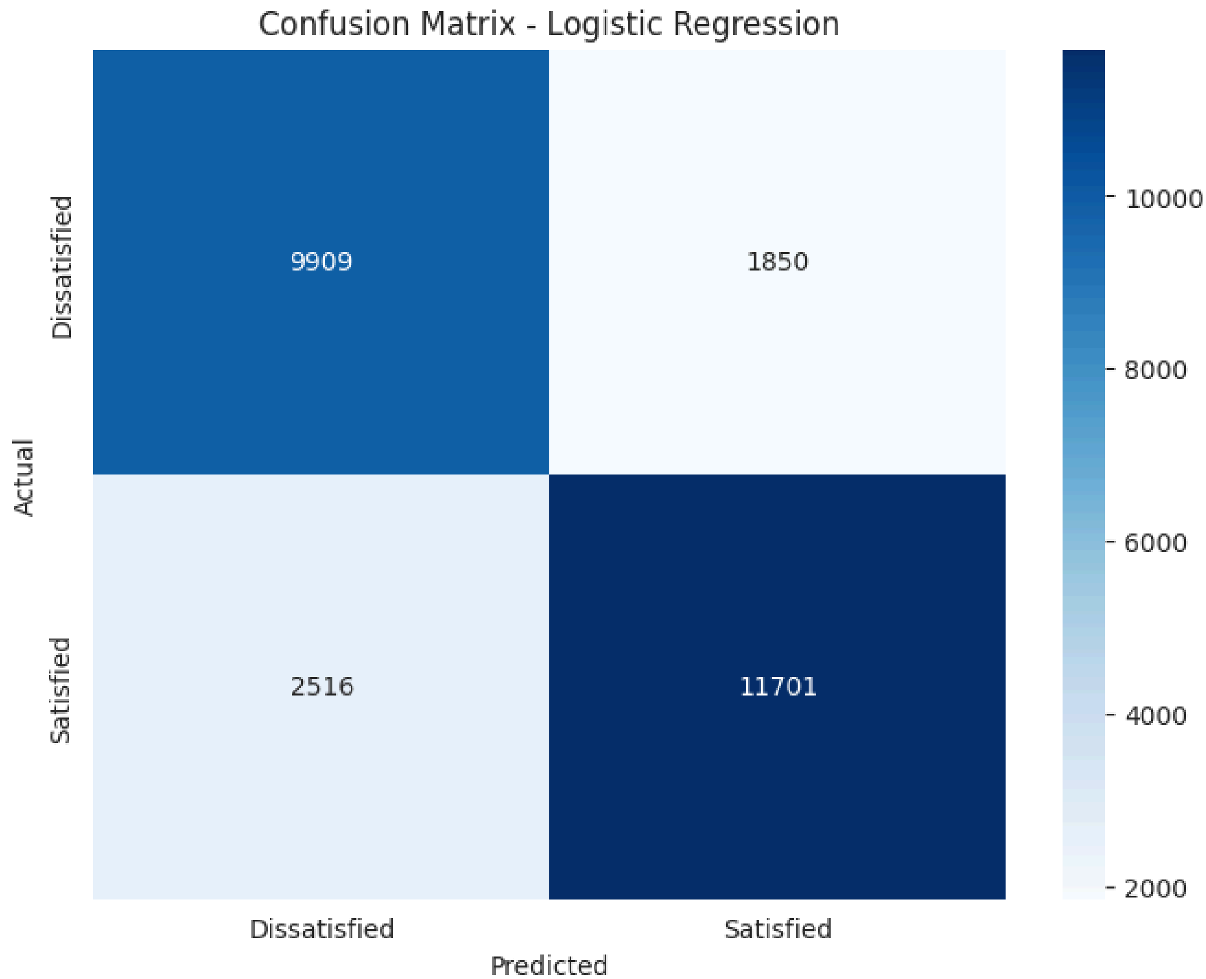
==================================================
MODEL PERFORMANCE METRICS
==================================================
Accuracy:  0.8319
Precision: 0.8635
Recall:    0.8230
F1-Score:  0.8428
ROC-AUC:   0.9032
==================================================

Confusion Matrix:
[[ 9909  1850]
 [ 2516 11701]]
```

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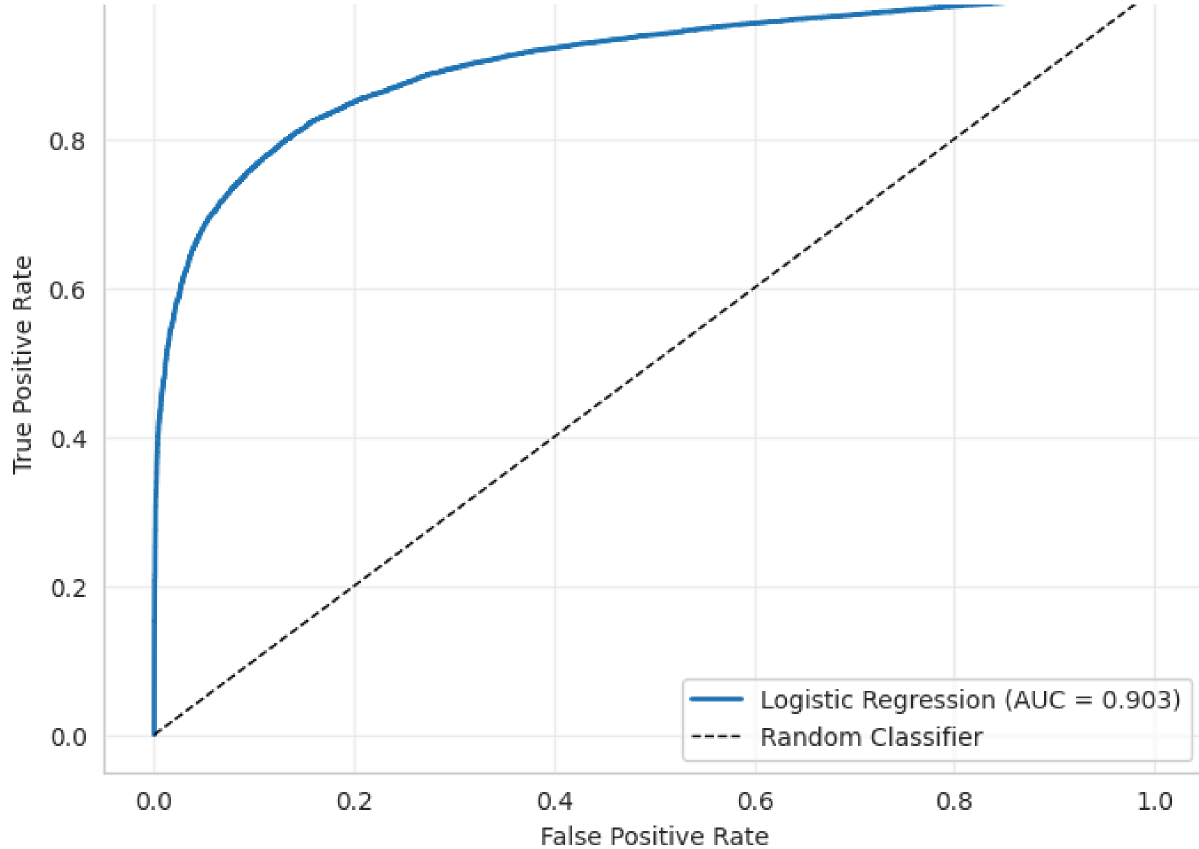
Classification Report:
              precision    recall  f1-score   support

Dissatisfied       0.80      0.84      0.82     11759
   Satisfied       0.86      0.82      0.84     14217

    accuracy                           0.83     25976
   macro avg       0.83      0.83      0.83     25976
weighted avg       0.83      0.83      0.83     25976
```



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TOP 10 FEATURES INFLUENCING SATISFACTION
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Feature	Coefficient	Odds_Ratio
Inflight entertainment	0.947759	2.579922
Customer Type	-0.763088	0.466224
Type of Travel	-0.429050	0.651127
Service_Quality_Avg	0.402527	1.495600
On-board service	0.337050	1.400809
Departure/Arrival time convenient	-0.332072	0.717436
Checkin service	0.300478	1.350504
Seat comfort	0.294630	1.342629
Class	-0.293338	0.745770
Food and drink	-0.277754	0.757483